SCN4A-Related Myopathy, Autosomal Recessive (MONDO:0100121): A Comprehensive Disease Characterization

Summary

SCN4A-Related Autosomal Recessive Myopathy (MONDO:0100121) is an ultra-rare congenital neuromuscular disorder caused by **biallelic loss-of-function (LOF) variants in *SCN4A***, the gene encoding the **skeletal-muscle voltage-gated sodium channel NaV1.4** (α-subunit, HGNC:10591; UniProt P35499; chromosome 17q23.3; gene MIM 603967). This is mechanistically the **mirror image** of the far more familiar dominant *SCN4A* channelopathies (hyperkalemic/hypokalemic periodic paralysis, paramyotonia congenita, sodium-channel myotonia), which arise from **gain-of-function or altered-gating** mutations. In the recessive disease, reduced or absent sodium current fails to depolarize the sarcolemma sufficiently to generate and propagate muscle action potentials, impairing excitation–contraction coupling and producing weakness rather than hyperexcitability.

The defining clinical feature is a **genotype–severity gradient governed by residual NaV1.4 current**. Complete biallelic loss (two null alleles) causes **lethal fetal akinesia/hypokinesia** — the most severe end, often resulting in intrauterine or neonatal death. A null allele combined with a hypomorphic (partial-function) allele, or two hypomorphic alleles, produces a **survivable congenital myopathy** with neonatal hypotonia, facial/neck/bulbar weakness, respiratory and feeding difficulty, and spinal deformity, which characteristically **improves over the first decade of life**. Specific partial-LOF missense variants (notably at residue **R1460**, and the historical **V1442E**) produce a **fatigable "myasthenic" congenital myopathy** that overlaps with congenital myasthenic syndrome (CMS16, OMIM 614198). The landmark cohort (Zaharieva et al., *Brain* 2016) found 7 of 11 affected individuals died in utero or shortly after birth, with the 4 survivors all improving clinically in childhood.

Converging lines of evidence confirm the recessive, dose-dependent logic: **gnomAD constraint** shows *SCN4A* tolerates heterozygous LOF (pLI ≈ 0, LOEUF 0.61), consistent with asymptomatic carriers; and a **NaV1.4-null mouse** recapitulates the human dose relationship (heterozygous nulls have latent myasthenia without periodic paralysis; homozygous nulls die by postnatal

day 2). Diagnosis is **molecular** — biallelic *SCN4A* variants detected by whole-exome/whole-genome sequencing or neuromuscular gene panels — because muscle biopsy shows only nonspecific myopathic changes and EMG shows a myopathic process without myotonia. **No disease-modifying therapy exists;** management is supportive (neonatal respiratory and feeding support, scoliosis management, physiotherapy), with inconsistent empirical benefit from salbutamol, acetazolamide, and acetylcholinesterase inhibitors tried by analogy to CMS.

Section 1 — Disease Information

Overview. SCN4A-Related Autosomal Recessive Myopathy is a Mendelian congenital neuromuscular disease caused by biallelic LOF variants in *SCN4A* (NaV1.4). It spans a phenotypic continuum from the most severe **congenital myopathy with severe fetal hypokinesia / fetal akinesia** (frequently lethal) through a **survivable congenital myopathy** (neonatal hypotonia, facial and bulbar weakness, respiratory compromise) to a **fatigable myasthenic congenital myopathy**. This entity is biologically and clinically distinct from the common autosomal **dominant** *SCN4A* channelopathies driven by gain-of-function/altered-gating mechanisms.

Key identifiers.

Resource	Identifier
Mondo	MONDO:0100121
Causal gene	*SCN4A* (HGNC:10591)
Related OMIM phenotype (myasthenic/CMS16)	614198 (congenital myasthenic syndrome 16)
SCN4A gene OMIM	603967
Ensembl	ENSG00000007314
NCBI Gene	6329
UniProt	P35499 (NaV1.4)
Cytogenetic locus	17q23.3

> Note: OMIM was recorded as "not recorded" in the disease template. The most closely linked OMIM phenotype for the myasthenic/recessive end of the spectrum is CMS16 (614198). The broader recessive congenital myopathy/fetal hypokinesia spectrum is captured under MONDO:0100121.

Synonyms / alternative names. - Congenital myopathy with "corona" fibers, selective muscle atrophy, and craniosynostosis - Congenital myopathy with severe fetal (foetal) hypokinesia - Myopathy with ptosis and mild dystrophic pattern - SCN4A-related congenital myopathy; recessive NaV1.4 loss-of-function myopathy

Information source. Derived from **aggregated disease-level resources** (OMIM, Orphanet, Mondo) and **individual-patient primary literature** — principally small case series and single case reports, reflecting the ultra-rarity of the condition (dozens of published patients worldwide).

Section 2 — Etiology

Primary cause — genetic. The disease is caused exclusively by **biallelic (homozygous or compound heterozygous) LOF variants in *SCN4A***. Zaharieva et al. identified such variants in 11 individuals from 6 unrelated kindreds; each affected case carried **at least one full loss-of-function (null) allele**, and 5 of 6 families had a second LOF allele in trans.

> *"we identified homozygous or compound heterozygous SCN4A mutations in a cohort of 11 individuals from six unrelated kindreds with congenital myopathy"* — PMID: 26700687

Genetic risk factors. The causal variants are the risk factor. Disease requires **two defective alleles**; a single LOF allele is insufficient (carriers are typically asymptomatic). **Consanguinity** increases risk by raising homozygosity. No modifier loci or susceptibility SNPs have been established; the principal "modifier" is the **identity of the second allele** (null vs hypomorph), which sets residual channel function and therefore severity.

Environmental risk factors. None established. This is a monogenic disorder; onset is congenital/prenatal, so postnatal environmental exposures are not causal. **Sex** is not a risk factor (no sex bias; the gene is autosomal). **Family history** / **consanguinity** are the only relevant demographic factors.

Protective factors. The principal "protective" factor is **residual NaV1.4 function** — a hypomorphic allele that retains partial current shifts the phenotype toward survivability and milder disease relative to a complete null. No protective modifier alleles or dietary/lifestyle protective factors are documented. At the population level, gnomAD shows heterozygous LOF is tolerated (pLI ≈ 0), so **a single functional allele is protective** against disease.

Gene–environment interactions. None established. Because manifestations are prenatal/congenital and fully penetrant with the biallelic genotype, there is no meaningful GxE axis documented.

Section 3 — Phenotypes

The phenotype is a continuum stratified by residual channel function. Key manifestations, with suggested HPO terms:

Phenotype	Type	Onset	Severity	Frequency	Suggested HPO
Fetal akinesia / severe hypokinesia	Clinical sign (prenatal)	Prenatal	Severe (often lethal)	Severe end (~7/11 in Zaharieva died in utero/ neonatally)	HP:0001558 (Decreased fetal movement)
Congenital hypotonia	Clinical sign	Neonatal	Moderate–severe	Survivors	HP:0001319 (Neonatal hypotonia)
Facial / neck muscle weakness	Clinical sign	Neonatal	Moderate	Survivors (characteristic)	HP:0030319 (Facial-muscle weakness); HP:0000467 (Neck-muscle weakness)
Bulbar weakness / swallowing difficulty	Clinical sign	Neonatal	Moderate–severe	Survivors	HP:0002015 (Dysphagia)
Respiratory insufficiency	Clinical sign	Neonatal	Severe	Common; drives mortality	HP:0002093 (Respiratory insufficiency)
Ptosis	Physical manifestation	Neonatal/ childhood	Mild–moderate	Subset ("myopathy with ptosis")	HP:0000508 (Ptosis)
Fatigable weakness (myasthenic)	Symptom	Neonatal/ childhood	Variable	Myasthenic subtype (R1460, V1442E)	HP:0003473 (Fatigable weakness)
Spinal deformity / scoliosis	Physical manifestation	Childhood	Moderate	Survivors	HP:0002650 (Scoliosis)
	Clinical sign				

Phenotype	Type	Onset	Severity	Frequency	Suggested HPO
Recurrent respiratory/ bulbar paralytic crises		Birth onward	Severe episodic	Myasthenic subtype (Tsujino)	HP:0002878 (Respiratory failure)
Arthrogryposis / contractures (fetal akinesia sequence)	Physical manifestation	Congenital	Variable	Severe end	HP:0002804 (Arthrogryposis multiplex congenita)

Progression. Severe fetal forms are lethal perinatally. **Survivors characteristically improve over the first decade of life** — an unusual and clinically important feature. Muscle biopsy in survivors shows nonspecific myopathic changes and EMG a myopathic process without myotonia (Zaharieva 2016).

> *"we identified homozygous or compound heterozygous SCN4A mutations in a cohort of 11 individuals from six unrelated kindreds with congenital myopathy"* — PMID: 26700687

Myasthenic subtype features. Fatigable generalized weakness with decremental CMAP on repetitive nerve stimulation and recurrent attacks of respiratory/bulbar paralysis since birth (Tsujino et al.).

> *"endplate potentials depolarizing the resting potential to -40 mV failed to excite action potentials"* — PMID: 12766226

Quality-of-life impact. Severe in the neonatal period (ventilator/feeding dependence, risk of death). In survivors, improvement over the first decade with residual facial/axial weakness, scoliosis, and fatigability; impact ranges from moderate disability to relatively preserved function depending on residual current.

Section 4 — Genetic / Molecular Information

Causal gene. *SCN4A* (NaV1.4), HGNC:10591, OMIM 603967, Ensembl ENSG00000007314, NCBI Gene 6329, chromosome 17q23.3. Encodes the pore-forming α-subunit of the adult skeletal-muscle voltage-gated sodium channel responsible for the rapid upstroke of the sarcolemmal action potential.

Pathogenic variants. Disease-causing alleles are **biallelic LOF**. Reported variant classes and examples:

Variant	Type	Functional consequence	Phenotype context	Source
Nonsense/null alleles (e.g., p.R1059X)	LOF (null)	Complete loss of function	Severe/lethal when biallelic; combines with hypomorph	Elia 2019 [PMID: 30824560]
p.R1460W (homozygous)	Missense, partial LOF	Reduced amplitude, hyperpolarized inactivation shift	Myasthenic congenital myopathy	[PMID: 30824560]
p.R1460Q (compound het with p.R1059X)	Missense, mixed LOF+GOF	Loss dominates; GOF features → myotonia in carriers	Myasthenic myopathy	[PMID: 30824560]
p.V1442E (with benign p.S246L in trans)	Missense	Enhanced fast inactivation near rest; enhanced use-dependent inactivation	Myasthenic syndrome (CMS16)	Tsujino 2003 [PMID: 12766226]
p.Arg1454Trp (LOF) + p.Asn1205Lys	Compound-het missense	LOF	Severe congenital myopathy	Berghold 2022 [PMID: 36090556]

Variant classification. Interpretation follows ACMG/AMP, but with a **mechanism-informed, phenotype-first workflow** because *SCN4A* produces directionally opposite mechanisms (dominant GOF vs recessive LOF). D'Ambrosio et al. emphasize integrating inheritance, channel topology, population frequency, segregation, and **mechanism-matched functional assays**.

> *"severe biallelic loss of function causing congenital myopathy and fetal hypokinesia"* — PMID: 42793010

Allele frequency / population constraint. gnomAD (GRCh38): **pLI = 3.99e-10 (≈0; NOT LOF-intolerant)**; LOF observed/expected o/e = 0.512 (90% CI 0.429–0.615; **LOEUF 0.61**); observed LOF = 83 vs expected 162; missense Z = 2.29 (mis o/e = 0.89). This pattern — tolerance of heterozygous LOF plus modest missense constraint — is exactly what is expected for a **recessive** disease gene where one working allele suffices.

Somatic vs germline. Entirely **germline**; no somatic involvement.

Functional consequence. Loss of function (reduced channel availability / current amplitude). This contrasts with dominant *SCN4A* disease (gain-of-function, altered gating, gating-pore leak currents). Some myasthenic-subtype missense variants (R1460) show **mixed** LOF+GOF, with LOF driving the recessive myasthenic phenotype.

> *"Recessive inheritance, with experimentally established loss-of-function, is a consistent feature of sodium channel based myasthenia, whereas the mixed gain of function for p.R1460 may also cause susceptibility to myotonia"* — PMID: 30824560

Modifier genes. None established; the trans allele is the primary determinant of residual function. **Epigenetic changes** and **chromosomal abnormalities**: none implicated — this is a point-mutation/small-variant disease.

Section 5 — Environmental Information

Environmental factors: None causal. **Lifestyle factors:** None causal (congenital/prenatal onset). **Infectious agents:** Not applicable — this is a monogenic channelopathy with no infectious trigger. The only non-genetic contributor to recurrence risk is **consanguinity** (a demographic, not environmental, factor).

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic LOF variants in *SCN4A*** (null/null, null/hypomorph, or hypomorph/hypomorph) **lead to** reduced or absent functional NaV1.4 α-subunit at the sarcolemma. *(Demonstrated: HEK293/Xenopus functional assays.)*
2. Reduced functional channel **results in** decreased macroscopic sodium current density (INa) in the muscle fiber membrane (sarcolemma, including t-tubules). *(Demonstrated in vitro and in NaV1.4-null mouse: INa ~50% of wild-type in heterozygotes, no compensation by fetal NaV1.5.)*
3. Decreased INa **leads to** failure to reach threshold / failure to generate and propagate muscle action potentials, especially during sustained high-frequency firing. *(Demonstrated: endplate potentials failed to excite action potentials in Tsujino 2003.)*
4. Action-potential failure **results in** impaired excitation–contraction coupling (insufficient t-tubular depolarization to trigger Cav1.1/RyR1 Ca^{2+} release). *(Inferred from channel physiology.)*
5. Impaired E–C coupling **leads to** muscle weakness. The clinical form then **branches by residual current:**
6. **5a. Complete biallelic null (≈0 residual current) → leads to** profound prenatal muscle inactivity → **fetal akinesia/hypokinesia sequence** (contractures, pulmonary hypoplasia) → **intrauterine or neonatal death.**
7. **5b. Null + hypomorph / biallelic hypomorph (low residual current) → leads to** survivable **congenital myopathy** (neonatal hypotonia, facial/bulbar/respiratory weakness, scoliosis) that **improves over the first decade** (inferred: postnatal developmental increase in channel expression/compensation).
8. **5c. Residue-specific partial LOF (e.g., R1460, V1442E; near-normal resting current but failure under load) → leads to** a **fatigable myasthenic congenital myopathy** with decremental CMAP and paralytic crises.

Detail by category

- **Biochemical abnormality / ion-channel defect (upstream, primary):** Reduced NaV1.4-mediated INa — a classic **sodium-channelopathy LOF** lesion. CHEBI:29101 (sodium ion).
- **Protein dysfunction:** Loss of function via reduced channel availability, hyperpolarized shift of fast inactivation (V1442E, R1460), enhanced use-dependent inactivation. For null

alleles, absent protein. GO suggestions: **GO:0005248** (voltage-gated sodium channel activity), **GO:0086010** (membrane depolarization during action potential), **GO:0001508** (action potential).

- **Cellular processes (downstream):** Failure of **skeletal muscle contraction** (GO:0003009) and **membrane depolarization** (GO:0051899); impaired E–C coupling. No apoptosis/inflammation signature is central; biopsy is non-inflammatory.
- **Tissue damage mechanism:** Primarily **functional** (electrical failure) rather than degenerative. Biopsy shows nonspecific myopathic changes (fiber-size variability, fibrofatty replacement of varying severity) — not a specific structural myopathy and not prominent necrosis.
- **Metabolic / immune involvement:** Not implicated. No autoimmunity (distinguishing it from acquired myasthenia gravis), no energy-metabolism defect (though one patient was initially misdiagnosed as mitochondrial myopathy).
- **Molecular profiling:** No disease-specific transcriptomic/proteomic/metabolomic signature established; mechanism is defined by electrophysiology.

Cell types (CL): skeletal muscle fiber / myofiber (**CL:0000188**, skeletal muscle cell). **Upstream vs downstream:** the ion-channel defect is upstream; E–C-coupling failure and weakness are downstream.

Section 7 — Anatomical Structures Affected

- **Organ level (primary):** Skeletal muscle (**UBERON:0001134**, skeletal muscle tissue; UBERON:0001630, muscle organ). Facial, neck/axial, bulbar, respiratory (diaphragm/intercostal), and limb muscles.
- **Secondary organ / system involvement:** Respiratory system (respiratory insufficiency/failure — the main driver of mortality; UBERON:0001004); skeletal system (scoliosis/spinal deformity, contractures; UBERON:0002090 axial skeleton). Fetal akinesia sequence secondarily affects lungs (hypoplasia) and joints.
- **Body systems: Musculoskeletal** (primary), **respiratory** (secondary). The **neuromuscular junction** is implicated functionally in the myasthenic subtype, but the defect is **post-synaptic / sarcolemmal** (muscle membrane), with normal ACh release and normal endplate structure in Tsujino's case.

- **Tissue/cell level:** Striated muscle tissue; **skeletal muscle fibers (CL:0000188)**. Epithelial/connective/nervous tissues are not primarily affected.
- **Subcellular level: Sarcolemma / plasma membrane** and **t-tubule system** where NaV1.4 resides. GO cellular component: **GO:0005886** (plasma membrane), **GO:0030315** (T-tubule).
- **Localization / lateralization: Bilateral and generalized** (symmetric), with facial, bulbar, axial, and respiratory predominance.

Section 8 — Temporal Development

- **Onset: Congenital / prenatal.** Severe forms present as reduced fetal movement (fetal akinesia/hypokinesia); survivable forms present with neonatal hypotonia. Myasthenic subtype: symptoms "since birth."
- **Onset pattern:** Insidious prenatal onset; myasthenic crises can be acute/episodic.
- **Progression / course:**
- *Severe (biallelic null):* lethal in utero or in the neonatal period.
- *Intermediate (survivable congenital myopathy):* **non-progressive to improving** — survivors improve over the **first decade of life**.
- *Myasthenic subtype:* **fluctuating/episodic** fatigable weakness with paralytic crises.
- **Disease duration:** Chronic lifelong in survivors; perinatally lethal at the severe end.
- **Remission pattern:** No true remission, but clinical **improvement** in childhood survivors (likely treatment-independent, developmental).
- **Critical periods:** Prenatal/perinatal period is the window of maximal vulnerability (akinesia sequence, respiratory failure). The neonatal period is the key intervention window (respiratory and feeding support determine survival).

Section 9 — Inheritance and Population

- **Inheritance: Autosomal recessive.** Requires two defective *SCN4A* alleles. This is the key contrast with dominant *SCN4A* channelopathies.
- **Penetrance:** Essentially **complete** for the biallelic genotype.

- **Expressivity: Highly variable**, determined by residual NaV1.4 current (null/null vs null/hypomorph vs hypomorph/hypomorph vs residue-specific partial LOF).
- **Carriers:** Obligate heterozygous carriers are **typically asymptomatic** (supported by gnomAD pLI ≈ 0). The NaV1.4-null mouse heterozygote shows only **latent (subclinical) myasthenia**.
- **Genetic anticipation:** Not applicable (no repeat expansion). **Germline mosaicism:** not specifically documented. **Founder effects:** none identified; variants are private across kindreds.
- **Consanguinity:** Contributory (several homozygous cases from consanguineous or isolated kindreds).
- **Epidemiology: Ultra-rare** — only dozens of published patients worldwide; formal prevalence/incidence not estimated.
- **Population demographics: Panethnic**, reported across multiple ethnicities with **no founder population**, **no sex bias** (autosomal), and onset at birth/prenatally.

> *"we identified homozygous or compound heterozygous SCN4A mutations in a cohort of 11 individuals from six unrelated kindreds with congenital myopathy"* — PMID: 26700687

Section 10 — Diagnostics

Diagnosis is molecular. Because clinical and histological features are nonspecific, definitive diagnosis requires detection of **biallelic *SCN4A* variants**.

- **Genetic testing (primary modality): Whole-exome sequencing (WES)** was the discovery method (Zaharieva 2016). **WGS**, **neuromuscular/congenital-myopathy/CMS gene panels** including *SCN4A*, and **single-gene testing** (when phenotype is strongly suggestive) are all applicable. Segregation analysis confirms biallelic/trans configuration. CMA, karyotyping, FISH, mtDNA, and repeat-expansion testing are **not relevant** (small-variant disease).

> *"severe biallelic loss of function causing congenital myopathy and fetal hypokinesia"* — PMID: 42793010

- **Muscle biopsy (supportive, nonspecific):** Myopathic features — increased fiber-size variability, fibrofatty tissue of varying severity — **without** specific structural abnormalities (no entity-specific cores/rods). Useful to exclude other congenital myopathies.
- **Electrophysiology:** EMG shows a **myopathic process without myotonia**. In the **myasthenic subtype**, repetitive nerve stimulation shows a **decremental CMAP**; microelectrode studies show endplate potentials that fail to excite action potentials despite normal ACh release and endplate morphology — localizing the defect to the **sarcolemma (post-synaptic muscle membrane)**.
- **Laboratory:** Creatine kinase usually normal or mildly elevated (nonspecific). No specific biomarker exists.
- **Functional assays (research/variant interpretation):** Voltage-clamp in HEK293/Xenopus oocytes to confirm LOF and classify variants per a mechanism-informed ACMG/AMP workflow.
- **Differential diagnosis:** Other congenital myopathies (*RYR1*, *ACTA1*, *NEB*, *TPM2/3*, *MTM1*), congenital myasthenic syndromes (CHRNE, RAPSN, DOK7, CHAT, AGRN, GFPT1, PLEC), SMA, congenital myotonic dystrophy, mitochondrial myopathy (one SCN4A patient was initially misdiagnosed as mitochondrial myopathy), and fetal akinesia deformation sequence of other causes.
- **Screening:** Not part of routine newborn screening. **Carrier and cascade screening** of relatives is appropriate once a familial variant is known; **prenatal/preimplantation testing** is available for at-risk couples.

Section 11 — Outcome / Prognosis

- **Survival / mortality: Bimodal and genotype-dependent.** Complete biallelic null → high perinatal mortality (**7 of 11 individuals in Zaharieva 2016 died in utero or shortly after birth**). Survivors (null+hypomorph / biallelic hypomorph) have a chronic but **improving** course.

- **Life expectancy:** Severe end — perinatally lethal. Survivable end — potentially near-normal with supportive care, though respiratory/bulbar weakness and scoliosis carry ongoing risk.
- **Morbidity / function:** Neonatal ventilator and feeding dependence; residual facial/axial weakness, fatigability, and scoliosis in survivors. Characteristic **improvement over the first decade** is a favorable prognostic feature.
- **Complications:** Respiratory failure (leading cause of death), aspiration from bulbar weakness, scoliosis/restrictive lung disease, contractures.
- **Prognostic factors:** The strongest predictor is **residual NaV1.4 current / genotype** (null/null worst; presence of a hypomorphic allele predicts survivability and improvement). Severity of neonatal respiratory compromise is the key clinical prognostic factor.

Section 12 — Treatment

No disease-modifying therapy exists. Management is **supportive and symptomatic**.

- **Supportive / rehabilitative (mainstay):** Neonatal **respiratory support** (non-invasive or invasive ventilation), **feeding/swallowing support** (NG/gastrostomy), **scoliosis/spinal deformity management**, and **physiotherapy/occupational therapy**. NCIT: Supportive Care (NCIT:C15329), Physical Therapy (NCIT:C15327), Mechanical Ventilation.
- **Empirical pharmacotherapy (inconsistent benefit):** By analogy to *SCN4A* congenital myasthenic syndrome, agents tried include **acetylcholinesterase inhibitors (pyridostigmine)**, **β2-agonists (salbutamol/albuterol)**, and **acetazolamide** — with **variable and often limited response**. In a genetically confirmed patient (compound heterozygous p.Arg1454Trp LOF + p.Asn1205Lys), there was limited benefit from salbutamol and acetazolamide, and pyridostigmine caused side effects at a minor dose.

> *"We present a now 18-year-old female patient with a severe congenital myopathy phenotype, originally diagnosed as mitochondrial myopathy, however later revealed to constitute a..."* — PMID: 36090556

- **Pharmacogenomics / personalized medicine:** Treatment is **genotype/mechanism-guided in principle** — the myasthenic subtype (partial-LOF, NMJ-like safety-margin failure) is the most plausible to respond to cholinergic augmentation, whereas pure structural LOF

myopathy is less likely to respond. This mirrors the general CMS principle that drugs beneficial in one subtype can be harmful in another.

- **Advanced therapeutics (gene/cell/RNA therapy), targeted therapy, immunotherapy:** None established or approved for this indication. No gene-replacement or ASO program is reported.
- **Surgical / interventional:** Scoliosis may require orthopedic surgical management.
- **Experimental:** No disease-specific registered clinical trials (no NCT identifiers specific to recessive SCN4A myopathy identified).

Section 13 — Prevention

- **Primary prevention:** Not possible for the congenital disease itself (prenatal onset). **Genetic counseling** for at-risk/consanguineous couples is central. For couples with a known familial biallelic risk, **preimplantation genetic testing (PGT)** and **prenatal diagnosis** allow reproductive risk management.
- **Secondary prevention: Carrier screening** and **cascade testing** of relatives once the familial variant is identified; early molecular diagnosis in a hypotonic neonate enables prompt respiratory/feeding support.
- **Tertiary prevention:** Proactive respiratory monitoring/ventilatory support, aspiration precautions, scoliosis surveillance and management, and physiotherapy to prevent contractures.
- **Immunization / public-health / environmental interventions:** Not applicable (non-infectious, monogenic).
- **Counseling:** Autosomal recessive counseling — 25% recurrence risk for carrier × carrier couples; emphasis on consanguinity-related risk.

Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs:** Human *SCN4A* (NCBI Gene 6329; NCBI Taxon 9606). Mouse ortholog **Scn4a** (NCBI Gene 110599; NCBI Taxon 10090) is well characterized. Orthologs are highly conserved across vertebrates, reflecting the essential role of NaV1.4 in skeletal-muscle excitability.

- **Natural disease in other species:** Dominant *SCN4A*-type channelopathies occur naturally in animals (e.g., equine hyperkalemic periodic paralysis — a gain-of-function phenotype). A naturally occurring **recessive LOF NaV1.4 myopathy** analogous to the human recessive disease is **not well documented** in companion animals/wildlife.
- **Comparative biology / evolutionary conservation:** The LOF mechanism is conserved — the mouse null phenotype (below) demonstrates that complete loss is neonatally lethal across mammals, confirming conserved essentiality.
- **Transmission:** Not applicable (non-infectious, non-zoonotic).

Section 15 — Model Organisms

- **Primary model — NaV1.4-null mouse (Wu et al., 2016):** Generated by deleting exon 12 of *Scn4a*. This genetic knockout **recapitulates the recessive, dose-dependent human genotype–phenotype relationship:**
- **Heterozygous nulls:** latent (subclinical) myasthenia with a right-shift of the force–stimulus relation; sodium current density ~50% of wild-type **without compensation by fetal NaV1.5**; **no periodic paralysis.**
- **Homozygous nulls: did not survive beyond the second postnatal day** — modeling human lethal fetal hypokinesia/complete biallelic null.

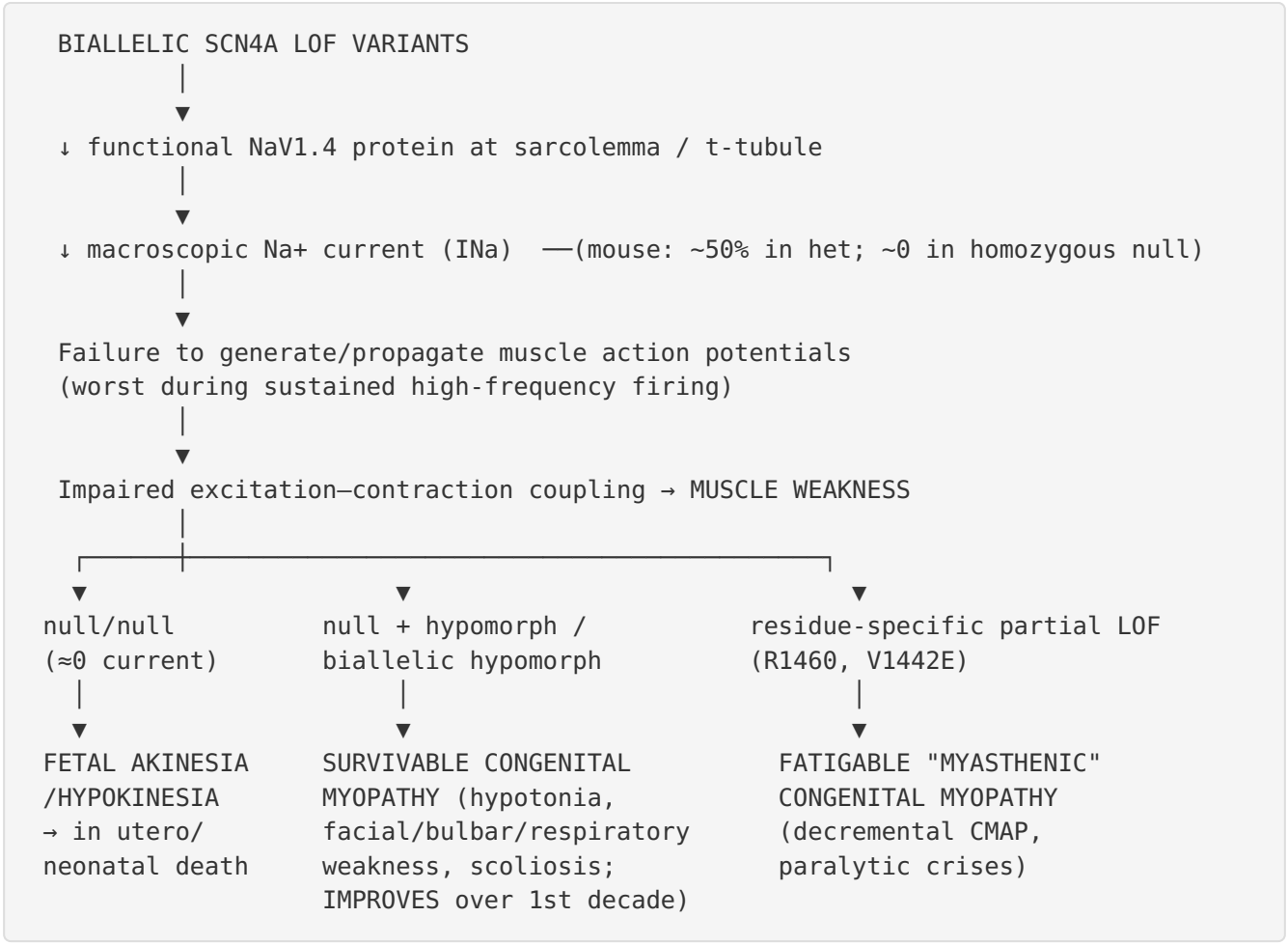
> *"Mice null for NaV1.4 did not survive beyond the second postnatal day"* — PMID: 27048647
>
> *"Heterozygous null mice have latent myasthenia and a right shift of the force-stimulus relation, without evidence of periodic paralysis"* — PMID: 27048647

- **In vitro / heterologous expression models: HEK293 cells** and **Xenopus oocytes** expressing mutant Nav1.4 channels, used for voltage-clamp characterization of LOF (reduced amplitude, hyperpolarized inactivation shifts, use-dependent inactivation) — the workhorse for variant interpretation (Tsujino 2003; Elia 2019).
- **Model type / genetic models:** Mammalian knockout (constitutive null); heterologous cell expression systems. Conditional/humanized/knock-in models specific to this disease are not prominently reported.

- **Phenotype recapitulation: Strong** — the mouse null reproduces both the carrier (latent myasthenia) and the lethal-homozygote ends of the human spectrum, directly validating the dose-dependent mechanism.
- **Limitations:** The homozygous null's early lethality limits study of the survivable/improving phenotype; hypomorphic knock-in models (null+hypomorph) that would mirror survivable human disease are not established. In vitro systems capture channel biophysics but not tissue-level E–C coupling or the developmental improvement seen in human survivors.
- **Resources:** MGI (mouse *Scn4a*), standard neuromuscular electrophysiology platforms.

Mechanistic Model / Interpretation

The unifying principle is a **continuous dose–response between residual NaV1.4 sodium current and clinical severity.** A single integrated model accounts for the entire phenotypic spectrum:



Residual current	Genotype	Phenotype	Outcome
≈ 0%	null / null	Fetal akinesia / severe hypokinesia	Perinatally lethal (7/11 in Zaharieva)
Low	null / hypomorph or hypomorph / hypomorph	Congenital myopathy, hypotonia, bulbar/respiratory weakness	Survivable; improves in first decade
Partial (load-dependent failure)	R1460W/W; R1460Q + null; V1442E	Fatigable myasthenic myopathy, crises	Chronic; fatigable
~50% (one null allele)	heterozygous carrier	Asymptomatic / latent myasthenia	No disease

This model is **mechanistically opposite** to dominant *SCN4A* disease, where gain-of-function or gating-pore leak currents cause hyperexcitability (myotonia) or depolarization-induced inexcitability (periodic paralysis). The recessive disease is a **pure loss-of-function hypoexcitability** disorder. Three independent evidence streams converge on this model: human genotype–phenotype series, in vitro electrophysiology confirming LOF, and the NaV1.4-null mouse reproducing both the carrier and lethal-null extremes. gnomAD constraint (pLI ≈ 0) independently confirms that heterozygous LOF is tolerated — the population-genetic signature of a recessive gene.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role
26700687	LOF *SCN4A* cause severe fetal hypokinesia or classical congenital myopathy (Zaharieva, *Brain* 2016)	Human cohort + in vitro	**Landmark**: establishes biallelic LOF → recessive congenital myopathy/fetal hypokinesia; 11 individuals/6 kindreds; 7/11 perinatal death; survivors improve; nonspecific biopsy
30824560	Myasthenic congenital myopathy from recessive R1460 mutations (Elia, *Neurology* 2019)	Human + voltage-clamp	Defines fatigable myasthenic subtype; recessive LOF consistent across sodium-channel myasthenia
12766226	Myasthenic syndrome V1442E (Tsujino, *PNAS* 2003)	Human + in vitro	First recessive *SCN4A* disorder (CMS16); sarcolemmal action-potential failure; normal ACh release/endplate
27048647	NaV1.4-null mice have latent myasthenia (Wu, 2016)	Mouse model	Validates dose-dependence: het = latent myasthenia; homozygous null = neonatal lethal
42793010	*SCN4A* channelopathies: mechanisms to variant interpretation (D'Ambrosio review)	Review	Frames recessive biallelic LOF vs dominant mechanisms; mechanism-informed ACMG/AMP workflow
36090556	Novel *SCN4A* case report (Berghold, 2022)	Human case	Genetically confirmed patient; documents limited/inconsistent pharmacotherapy response; misdiagnosis as mitochondrial myopathy
36835142, 29478601, 22353287, 15367858	CMS reviews	Reviews	Context: *SCN4A* as a recognized CMS gene; supportive pharmacology by analogy

PMID	Title (abbrev.)	Evidence type	Role
gnomAD (ENSG00000007314)	Constraint metrics	Population genetics	pLI ≈ 0, LOEUF 0.61 → heterozygous LOF tolerated (recessive logic)

Consistency: All four primary human/model studies and the population-constraint data point in the same direction — **recessive loss of function with severity set by residual current.** No contradictory evidence was found; the dominant *SCN4A* literature is complementary, not contradictory, because it describes a different (gain-of-function) mechanism.

Limitations and Knowledge Gaps

1. **Ultra-rarity and small samples.** Conclusions rest on a handful of case series/reports (dozens of patients total). No formal prevalence/incidence, natural-history cohort, or registry exists.
2. **No validated biomarker or quantitative severity scale** linking residual INa to clinical outcome in humans; the dose–response model is strongly supported but not quantitatively calibrated in patients.
3. **No survivable-phenotype animal model.** The homozygous null mouse dies neonatally; a hypomorphic/knock-in mouse that would model the survivable, improving human phenotype is lacking, limiting preclinical therapeutic testing.
4. **Mechanism of childhood improvement is unexplained** (hypothesized developmental increase in channel expression or compensatory remodeling — not demonstrated).
5. **Therapeutics are empirical.** No controlled trials; responses to salbutamol/acetazolamide/pyridostigmine are anecdotal and inconsistent. No genotype-stratified treatment evidence.
6. **Variant interpretation is challenging** because *SCN4A* produces opposite mechanisms; functional assays are needed but not universally available.
7. **OMIM entity boundaries** for the recessive congenital-myopathy/fetal-hypokinesia end vs CMS16 (614198) are not fully harmonized.

Proposed Follow-up Experiments / Actions

1. **International natural-history registry** for biallelic *SCN4A* patients to quantify prevalence, survival, and the trajectory of childhood improvement, stratified by genotype (null/null vs null/hypomorph vs hypomorph/hypomorph vs R1460-type).
2. **Genotype–current–phenotype calibration:** systematically voltage-clamp all reported recessive variants under a standardized protocol to derive a quantitative "residual-current index" and correlate with clinical severity.
3. **Hypomorphic/knock-in mouse models** (e.g., R1460W or patient hypomorphic alleles) to model the survivable, improving phenotype and serve as a therapeutic testing platform.
4. **Developmental expression study** of NaV1.4 (and any compensatory channels) in survivors or model muscle to explain childhood improvement — testing the developmental-compensation hypothesis.
5. **Genotype-stratified therapeutic trials/n-of-1 protocols** for the myasthenic subtype (AChE inhibitors, β2-agonists, acetazolamide, 3,4-diaminopyridine), prioritizing partial-LOF genotypes most likely to benefit.
6. **Explore current-enhancing strategies** (e.g., agents that reduce fast inactivation or increase channel availability) specifically for partial-LOF variants with hyperpolarized inactivation shifts (R1460, V1442E).
7. **Standardize variant-curation guidance** (ClinGen/ACMG mechanism-informed framework) distinguishing recessive LOF from dominant GOF *SCN4A* alleles, to reduce misclassification.

Ontology Term Appendix

- **Disease:** MONDO:0100121
- **Gene/Protein:** *SCN4A* HGNC:10591; UniProt P35499 (NaV1.4); NCBI Gene 6329
- **HPO (selected):** HP:0001558 (decreased fetal movement), HP:0001319 (neonatal hypotonia), HP:0030319 (facial-muscle weakness), HP:0002015 (dysphagia), HP:0002093 (respiratory insufficiency), HP:0000508 (ptosis), HP:0003473 (fatigable weakness), HP:0002650 (scoliosis), HP:0002804 (arthrogryposis multiplex congenita)
- **GO (BP):** GO:0001508 (action potential), GO:0086010 (membrane depolarization during AP), GO:0003009 (skeletal muscle contraction), GO:0005248 (voltage-gated sodium channel activity)

- **GO (CC):** GO:0005886 (plasma membrane), GO:0030315 (T-tubule)
- **CL:** CL:0000188 (skeletal muscle cell / myofiber)
- **UBERON:** UBERON:0001134 (skeletal muscle tissue), UBERON:0001004 (respiratory system), UBERON:0002090 (axial skeleton)
- **CHEBI:** CHEBI:29101 (sodium ion)
- **NCIT (interventions):** supportive care (C15329), mechanical ventilation, physical therapy (C15327), pyridostigmine, salbutamol, acetazolamide

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Report compiled from a 5-iteration autonomous investigation integrating human clinical cohorts, in vitro electrophysiology, a mouse knockout model, population-genetic constraint (gnomAD), and review literature. Evidence types are indicated throughout; all mechanistic and clinical claims are anchored to cited PMIDs.
